



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Phenomenology of the Standard Model of Elementary Particles

Master's Degree in Physics

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Table of Contents

1 Introduction

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



Probing the EW Sector and BSM Physics with $W^\pm Zjj$

1 Introduction

Experimental Setup & Target

- **Dataset:** 140 fb^{-1} of pp collisions at $\sqrt{s} = 13 \text{ TeV}$ (2015–2018).
- **Target:** Fully leptonic decays of $W^\pm Zjj$ production.

Analysis Objectives

- **Measurements:** Integrated & differential fiducial cross-sections (σ_{EW} and σ_{strong}).
- **Goal:** Isolate Vector Boson Scattering (VBS).

New Physics Search

- **BSM:** EFT limits on Anomalous Quartic Gauge Couplings (aQGC).



The ATLAS Detector

1 Introduction

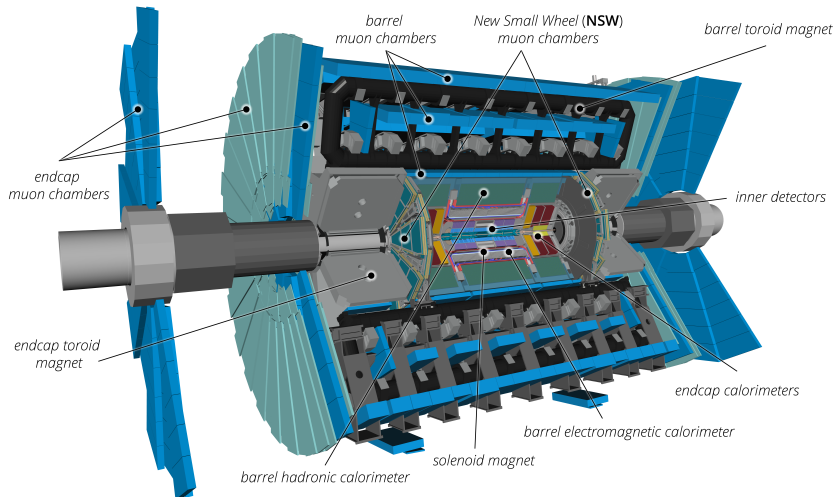




Table of Contents

2 Theoretical Framework

- ▶ Introduction
- ▶ **Theoretical Framework**
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



Gauge Boson Self-Interactions

2 Theoretical Framework

- The Electroweak (EW) sector of the Standard Model is based on the **non-abelian** gauge symmetry $SU(2)_L \times U(1)_Y$.
- Unlike QED, the generators of $SU(2)_L$ do not commute. As a consequence, the physical vector bosons ($W^\pm, Z, \gamma$) **self-interact**.
- This structure directly dictates the existence of **Triple** (TGC) and **Quartic** (QGC) Gauge Couplings:

$$\mathcal{L}_{\text{TGC}} = \frac{1}{2} g \epsilon_{abc} W_b^\mu W_c^\nu (\partial_\mu W_{a\nu} - \partial_\nu W_{a\mu}) \quad \mathcal{L}_{\text{QGC}} = \frac{1}{4} g^2 \epsilon_{abc} \epsilon_{ars} W_b^\mu W_c^\nu W_{r\mu} W_{s\nu}$$

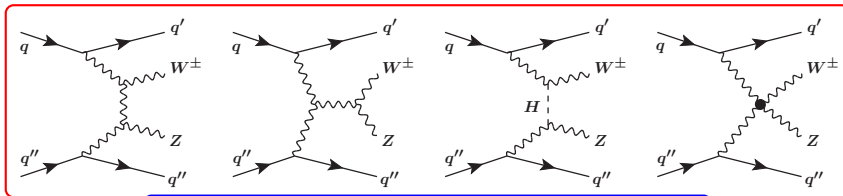
Probe these vertices to test the SM rigidity and search for anomalous couplings (aQGC) arising from **New Physics**.



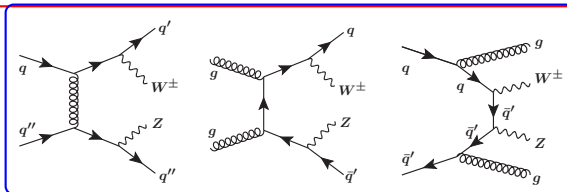
Experimental Signature: Vector Boson Scattering

2 Theoretical Framework

Vector Boson Scattering (VBS) $VV \rightarrow VV$ with $V = W/Z/\gamma$, is a key process with which to probe the $SU(2)_L \times U(1)_Y$ gauge symmetry of the **electroweak** (EW) theory that determines the **self-couplings** of vector bosons. To study these couplings, we exploit the purely electroweak production of $W^\pm Zjj$.



$W^\pm Zjj$ -EW



$W^\pm Zjj$ -QCD



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The $W^\pm Zjj$ -EW production can be sensitive to effects beyond the SM affecting the quartic interactions of weak bosons. The measured $W^\pm Zjj$ events are therefore used to search for anomalous quartic gauge couplings (aQGC) using an **EFT framework**. In this model, the Lagrangian of the standard model is extended into an efficient Lagrangian by adding higher-order operators and their respective **Wilson coefficients** as follows:

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\text{SM}} + \sum_i \frac{f_i^{(6)}}{\Lambda^2} O_i^{(6)} + \sum_j \frac{f_j^{(8)}}{\Lambda^4} O_j^{(8)} + \dots$$

Where:

- $O_{i,j}^{(6),(8)}$: are dimension-6 and dimension-8 operators;
- $f_i^{(6)}$ e $f_j^{(8)}$: dimensionless couplings
- Λ : is the energy scale of the new processes



A little look at the Effective Field Theory (EFT)

2 Theoretical Framework

The squared scattering amplitude of the effective field theory prediction for $W^\pm Z jj$ production can be written as:

$$\left| A_{\text{SM}} + \sum_i c_i A_i \right|^2 = |A_{\text{SM}}|^2 + \sum_i c_i 2 \text{Re}(A_{\text{SM}}^* A_i) + \sum_i c_i^2 |A_i|^2 + \sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$$

- $c_i = f_j^{(8)} / \Lambda^4$, A_{SM} is the SM scattering amplitude;
- $2 \text{Re}(A_{\text{SM}}^* A_i)$ is the amplitude of the interference term between the SM and the dimension-8 operators;
- $\sum_i c_i^2 |A_i|^2$ is the pure dimension-8 contribution, and $\sum_{ij, i \neq j} c_i c_j 2 \text{Re}(A_i A_j^*)$ is the amplitude of interferences between two dimension-8 operators, called cross terms.



Table of Contents

3 Fiducial Phase Space and MC Simulations

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ **Fiducial Phase Space and MC Simulations**
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



Fiducial Phase Space for Leptons

3 Fiducial Phase Space and MC Simulations

The $W^\pm Z jj$ cross-sections are measured in a **fiducial phase space**. Leptons produced in the decay of a hadron, a τ -lepton, or their descendants are not considered in the definition of the fiducial phase space and are treated as background.

Category	Kinematic Variable	Requirement
Z leptons	Transverse momentum ($p_T^{\ell_Z}$)	$> 15 \text{ GeV}$
W lepton	Transverse momentum ($p_T^{\ell_W}$)	$> 20 \text{ GeV}$
All leptons	Pseudorapidity ($ \eta_\ell $)	< 2.5
Z boson	Invariant mass window ($ m_{\ell^+\ell^-} - m_Z^{\text{PDG}} $)	$< 10 \text{ GeV}$
W boson	Transverse mass (m_T^W)	$> 30 \text{ GeV}$
Angular isolation	Distance between Z leptons ($\Delta R(\ell_{Z1}, \ell_{Z2})$)	> 0.2
Angular isolation	Distance between W and Z leptons ($\Delta R(\ell_W, \ell_Z)$)	> 0.3



Fiducial Phase Space for Jets

3 Fiducial Phase Space and MC Simulations

In addition, at least **two jets are required**, reconstructed using the anti- k_t algorithm with radius parameter $R = 0.4$. Hard processes with a b -quark in the initial or final states, such as tZj production, are not considered as part of the $W^\pm Zjj$ -EW events.

Category	Kinematic Variable	Requirement
All jets	Transverse momentum (p_T^j)	$> 40 \text{ GeV}$
All jets	Pseudorapidity ($ \eta_j $)	< 4.5
Angular isolation	Distance between jets and leptons ($\Delta R(j, \ell)$)	> 0.3
Tagging jets	Opposite hemispheres ($\eta_{j1} \cdot \eta_{j2}$)	< 0
Tagging jets	Invariant mass (m_{jj})	$> 500 \text{ GeV}$
Jet Veto		
b -quark jets	Absence of b -jets in initial/final state	$N_{b\text{-jet}} = 0$



How to eliminate leptons ambiguity?

3 Fiducial Phase Space and MC Simulations

- Leptons and final-state neutrons that do not originate from hadron or τ -lepton decays, are matched to the $W^\pm$ and Z boson decay products using an algorithmic approach, called the **resonant shape algorithm**.
- This algorithm is based on the value of an estimator expressing the product of the nominal line-shapes of the W and Z resonances:

$$P = \left| \frac{1}{m_{(\ell^+, \ell^-)}^2 - (m_Z^{\text{PDG}})^2 + i \Gamma_Z^{\text{PDG}} m_Z^{\text{PDG}}} \right|^2 \times \left| \frac{1}{m_{(\ell', \nu_{\ell'})}^2 - (m_W^{\text{PDG}})^2 + i \Gamma_W^{\text{PDG}} m_W^{\text{PDG}}} \right|^2$$



Monte Carlo Simulation of the Signal

3 Fiducial Phase Space and MC Simulations

The simulation of $W^{\pm}Zjj$ events was primarily generated using the MadGraph5_aMC@NL02.6.5 MC event generator, interfaced with Pythia8.240. The simulation was divided into distinct components:

- **Electroweak production $W^{\pm}Zjj$ – EW:** The actual signal. It is calculated at leading order (LO), considering purely electroweak processes (order α_{EW}^6).
- **QCD production $W^{\pm}Zjj$ – QCD:** Considered a background for the VBS analysis. This component is calculated at next-to-leading order (NLO) in QCD.
- **Interference $W^{\pm}Zjj$ – INT:** Quantifies the quantum interference between the EW and QCD production. It is calculated at LO and yields a positive contribution of 6% in the fiducial region, reaching up to 12% for events with three or more jets.



Background and Detector Simulation

3 Fiducial Phase Space and MC Simulations

In addition to the $W^{\pm}Zjj$ – QCD production, there are other processes that can mimic the signal in the detector.

- **Backgrounds with *prompt* (real) leptons:** Processes such as the production of ZZ pairs, WW , tribosons (VVV), or top quarks associated with vector bosons ($t\bar{t}V$, tZj). Their contribution is evaluated entirely using Monte Carlo simulations.
- **Backgrounds with *fake* leptons:** Events where a hadronic jet is misidentified as a lepton by the detector. Estimated directly using data-driven techniques.

All generated MC events were passed through the ATLAS detector simulation, based on Geant4, and processed using the same reconstruction software used for the data.



Table of Contents

4 Event selection & Background estimation

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ **Event selection & Background estimation**
- ▶ Measurement methodology and results
- ▶ Conclusion



Event Selection: Physics Objects Definition

4 Event selection & Background estimation

Data Quality & Trigger:

- Only data with stable beams and fully operational sub-detectors.
- Single-lepton triggers used (Combined efficiency $\approx 99\%$ for offline-selected $W^\pm Zjj$ events).

Physics Objects Reconstruction:

- **Prompt Leptons** e, μ, ν (E_T^{miss}): Reconstructed combining ID tracks with ECAL clusters or MS tracks. Strict isolation and impact parameter criteria are applied to reject non-prompt leptons.
- **Hadronic Jets**: Clustered using the *anti- k_t* algorithm ($R = 0.4$).
- **b-tagging**: Deep-learning algorithms identify displaced vertices from b-hadron decays (crucial for top-quark veto).



Physics Objects: Leptons

4 Event selection & Background estimation

Object	Selection Criteria
Muons	<i>Medium</i> Identification (Inner Detector + Muon Spectrometer) $p_T > 15 \text{ GeV}$, $ \eta < 2.5$ Impact parameters: $ d_0/\sigma_{d_0} < 3.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 90 – 99% (depending on p_T)
Electrons	<i>Medium Likelihood</i> Identification $p_T > 15 \text{ GeV}$, $ \eta < 1.37$ or $1.52 < \eta < 2.47$ (crack region veto) Impact parameters: $ d_0/\sigma_{d_0} < 5.0$, $ z_0 \sin \theta < 0.5 \text{ mm}$ Isolation efficiency: 89 – 99% (depending on p_T)

*Note: Impact parameter cuts are crucial to select **prompt** leptons and reject non-prompt background from heavy-flavor decays.*



Physics Objects: Jets & E_T^{miss}

4 Event selection & Background estimation

Object		Selection Criteria
Jets		Clustered via <i>anti-k_t</i> algorithm ($R = 0.4$) $p_T > 25 \text{ GeV}$, $ \eta < 4.5$ Pile-up suppression (JVT): <i>tight</i> ($p_T < 60 \text{ GeV}$, $ \eta < 2.4$) and <i>loose</i> ($p_T < 120 \text{ GeV}$, $2.5 < \eta < 4.5$)
Missing (E_T^{miss})	Energy	Proxy for the undetected neutrino. Computed as the negative vector sum of all calibrated <i>hard</i> objects (leptons, jets) plus a track-based <i>soft</i> term.



Event Selection: Boson Reconstruction

4 Event selection & Background estimation

Category	Selection Criteria
Baseline	Exactly 3 lepton candidates (μ or e) Veto on a 4th <i>loose</i> lepton ($p_T > 5$ GeV) Trigger matching: at least 1 lepton with $p_T > 25/27$ GeV
Z Boson Candidate	Same-Flavor Opposite-Sign lepton pair $ m_{\ell\ell} - m_Z^{\text{PDG}} < 10$ GeV (If ambiguous, the pair closest to m_Z is chosen)
W Boson Candidate	The remaining 3rd lepton + E_T^{miss} Stricter requirements: <i>Tight</i> ID & Isolation, $p_T > 20$ GeV Transverse mass: $m_T^W > 30$ GeV



Irreducible Backgrounds

4 Event selection & Background estimation

Irreducible backgrounds feature true prompt leptons mimicking the signal topology.

ZZ Control Region (ZZ-CR)

- **Background:** $ZZ \rightarrow 4\ell$ (ZZjj-QCD).
- **Origin:** Survives via lepton acceptance loss.
- **Selection:** Requires a 4th *loose* lepton to ensure orthogonality.

b Control Region (b-CR)

- **Background:** Top-quark ($t\bar{t} + V, tZj$).
- **Origin:** Survives via b-tagging inefficiency.
- **Selection:** Inverts the b-veto by requiring ≥ 1 b-jet.

While rare processes like **Tribosons** (VVV) and **ZZjj-EW** are estimated via **pure Monte Carlo** due to their low yields, the dominant sources are constrained using dedicated **Control Regions**:



Reducible Backgrounds (Fake Leptons)

4 Event selection & Background estimation

Unlike irreducible processes, reducible backgrounds in the $W^{\pm}Zjj$ phase space arise from instrumental misidentification rather than true prompt leptons.

Fake E_T^{miss} & Misidentified Objects

- **$Z + \text{jets}$:** A hadronic jet (e.g., charged pion) is misidentified as a lepton. E_T^{miss} is faked by finite Jet Energy Resolution.
- **$Z\gamma$:** A true photon mimics an electron.

True E_T^{miss} & Non-Prompt Leptons

- **$t\bar{t}$ & Wt :** True E_T^{miss} from $W \rightarrow \ell\nu$. The 3rd lepton is a non-prompt decay from a b -hadron inside a jet.
- **WW :** Two prompt leptons and true neutrinos. The 3rd lepton is faked by associated QCD jets or pile-up tracks.

Estimation Strategy: Data-Driven Matrix Method



Expected and Observed Numbers of Events Pre-Fit

4 Event selection & Background estimation

	SR ($N_{\text{jet}} = 2$)	SR ($N_{\text{jet}} \geq 3$)	b -CR	ZZ-CR
Data	169	477	666	210
Total pred.	231 ± 12	550 ± 50	660 ± 40	205 ± 11
$W^{\pm}Zjj$ — EW	65.0 ± 3.5	60 ± 6	4.82 ± 0.28	0.725 ± 0.014
$W^{\pm}Zjj$ — QCD	125 ± 9	380 ± 50	77 ± 18	6.2 ± 0.7
$W^{\pm}Zjj$ -INT	1.3 ± 0.6	5.3 ± 2.6	0.58 ± 0.29	0.22 ± 0.11
$t\bar{t} + V$	0.66 ± 0.04	20.2 ± 0.7	289 ± 10	9.89 ± 0.28
tZj	8.78 ± 0.34	19.7 ± 1.2	134 ± 4	0.432 ± 0.005
ZZ-QCD	9.6 ± 0.4	32.0 ± 2.5	10.1 ± 0.6	159 ± 9
ZZ-EW	2.2 ± 0.6	4.4 ± 1.1	0.25 ± 0.06	23 ± 6
VVV	0.41 ± 0.10	2.0 ± 0.5	0.39 ± 0.10	4.1 ± 1.1
Misid. leptons	18 ± 4	28 ± 7	150 ± 40	1.7 ± 0.5



Table of Contents

5 Measurement methodology and results

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ **Measurement methodology and results**
- ▶ Conclusion



Signal Extraction: Multivariate Analysis (BDT)

5 Measurement methodology and results

The Challenge

$WZjj$ -EW (signal, $\mathcal{O}(\alpha_{EW}^6)$) and $WZjj$ -QCD (background, $\mathcal{O}(\alpha_s^2 \alpha_{EW}^4)$) share the exact same final state. Kinematic cut-based separation is sub-optimal.

⇒ **Solution:** Boosted Decision Tree (TMVA).

BDT Input Variables (15 observables):

- *Jet*: m_{jj} , $|\Delta\eta_{jj}|$, $\Delta\phi_{jj}$, N_{jets}
- *Boson*: p_T^W , p_T^Z , m_T^{WZ}
- *Correlations*: Lepton centrality, event balance

BDT Output:

- Maps a multi-dimensional phase space into a **single discriminant score**.
- Used as the primary observable for signal extraction.

Note: An independent auxiliary BDT is deployed in the b -CR to differentiate between $t\bar{t}V$ and tZj .



The Global Likelihood Fit

5 Measurement methodology and results

Simultaneous Extended Binned Negative Log-Likelihood:

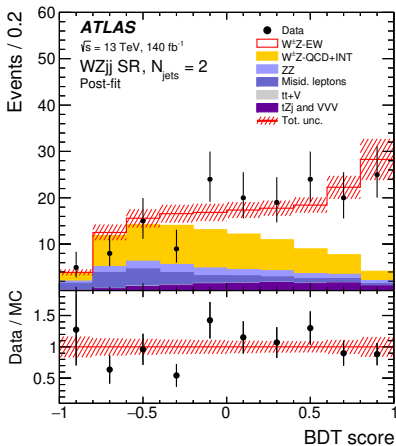
$$-\ln \mathcal{L}(\boldsymbol{\mu}, \mathbf{K}, \boldsymbol{\nu}) = \sum_r \sum_i \left(\mu_{\text{exp}}^{[r,i]} - N^{[r,i]} \ln \mu_{\text{exp}}^{[r,i]} \right) + \sum_j \frac{\nu_j^2}{2}$$

- **Regions (r):** SR ($N_{\text{jets}} = 2$), SR ($N_{\text{jets}} \geq 3$), b -CR, ZZ-CR.
- **Observables ($N^{[r,i]}$):** bins BDT score in SRs and in b -CR and m_{jj} in ZZ-CR.
- **Parameters of Interest ($\boldsymbol{\mu}$):** Strength modifiers μ_{EW} and μ_{QCD} .
- **Unconstrained Parameters ($\mathbf{K}$):** Background normalizations ($K_{t\bar{t}V}$, K_{tZj} , K_{ZZ}) extracted dynamically from CRs.
- **Nuisance Parameters ($\boldsymbol{\nu}$):** Theoretical/experimental systematics, constrained by the Gaussian penalty term $\nu_j^2/2$.

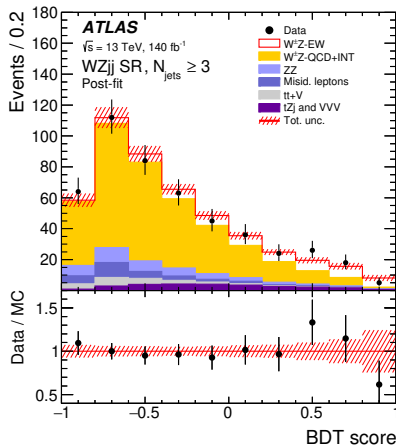


Post-fit distributions SR ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

5 Measurement methodology and results



(a)

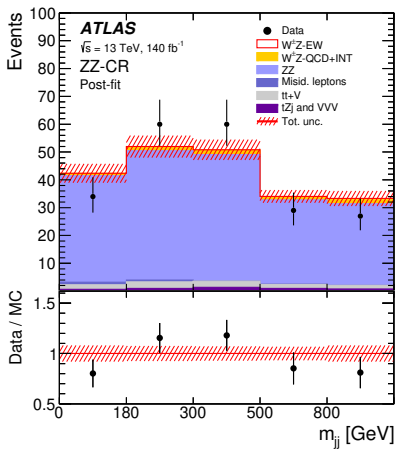


(b)

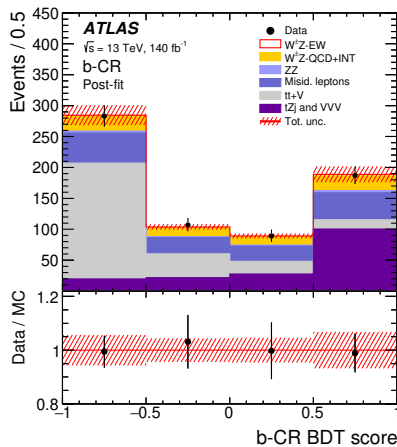


Post-fit distributions ZZ-CR, b-CR

5 Measurement methodology and results



(c)



(d)



Fiducial Cross-Section Extraction

5 Measurement methodology and results

The fiducial cross-sections are extracted directly from the profile-likelihood fit. The normalisation factors (μ_{EW} , μ_{QCD}) scale the theoretical MC predictions within the fiducial phase space:

$$\sigma_{WZjj-EW} = \sum_i \mu_i^{WZjj-EW} \cdot \sigma_{i,th.MC}^{WZjj-EW}$$
$$\sigma_{WZjj-strong} = \sum_i \left(\mu_i^{WZjj-QCD} \cdot \sigma_{i,th.MC}^{WZjj-QCD} + \mu_i^{WZjj-INT} \cdot \sigma_{i,th.MC}^{WZjj-INT} \right)$$

*Index i runs over the two SR categories: $N_{jets} = 2$ and $N_{jets} \geq 3$.

- **Interference term:** μ_{INT} is constrained as $\sqrt{\mu_{EW}} \cdot \sqrt{\mu_{QCD}}$.



Results and Comparison with SM Predictions

5 Measurement methodology and results

VBS (EW Production)

$$\sigma_{\text{obs}}^{\text{EW}} = 0.368 \pm 0.037_{\text{stat}} \pm 0.059_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{EW}} = 0.370 \pm 0.030 \text{ fb}$$

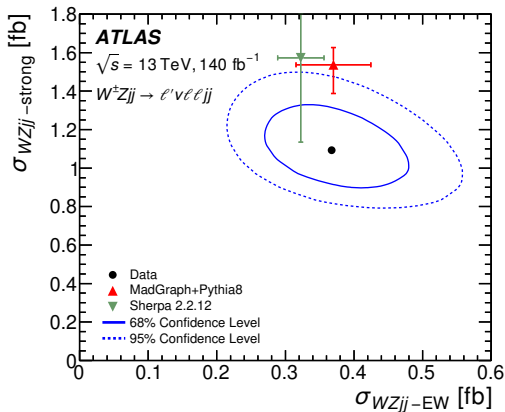
⇒ Excellent SM agreement

Strong + Int.

$$\sigma_{\text{obs}}^{\text{strong}} = 1.093 \pm 0.066_{\text{stat}} \pm 0.131_{\text{syst}} \text{ fb}$$

$$\sigma_{\text{th}}^{\text{strong}} = 1.537 \pm 0.150 \text{ fb}$$

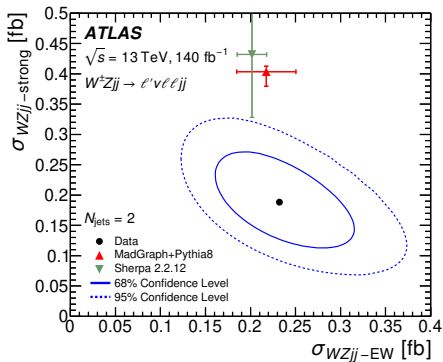
⇒ Deficit factor ~ 0.7 (1.8σ tension)



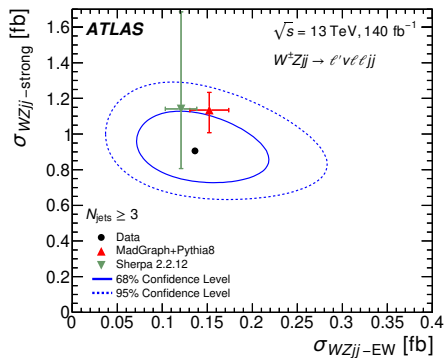


$W^{\pm}Zjj$ – EW and $W^{\pm}Zjj$ -strong ($N_{jets} = 2$), SR ($N_{jets} \geq 3$)

5 Measurement methodology and results



(a)

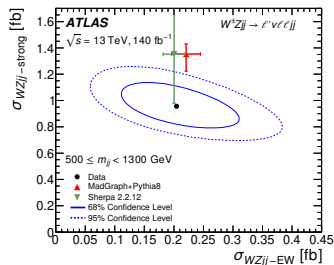


(b)

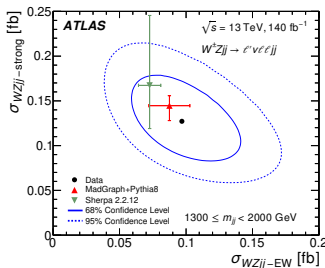


$W^{\pm}Zjj$ – EW and $W^{\pm}Zjj$ -strong in different m_{jj} bins

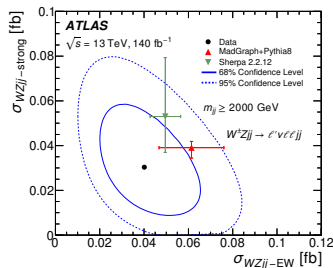
5 Measurement methodology and results



(a)



(b)



(c)



$W^\pm Z jj$ differential measurements

5 Measurement methodology and results

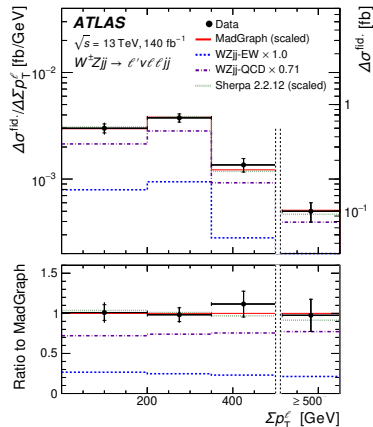
Differential measurements of $W^\pm Z jj$ production cross-section are performed as a function of variables sensitive to anomalies in the quartic gauge coupling.

Kinematic Variables

$\sum \mathbf{p}_T^\ell$ Scalar sum of the transverse momenta of the three charged leptons (W , Z).

$\Delta\phi(W, Z)$ Difference in azimuthal angle between the W and Z bosons' directions.

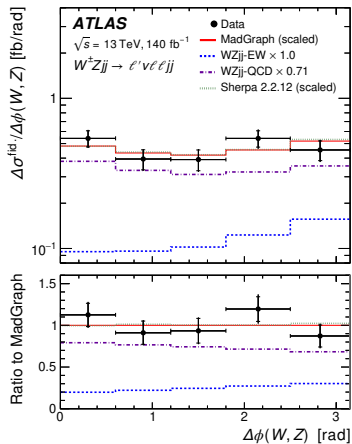
m_T^{WZ} Transverse mass of the $W^\pm Z$ system.



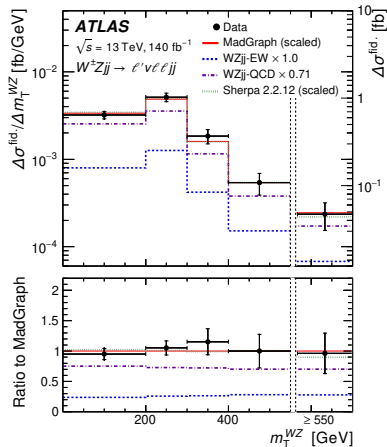


$W^{\pm}Zjj$ differential measurements

5 Measurement methodology and results



(a)



(b)



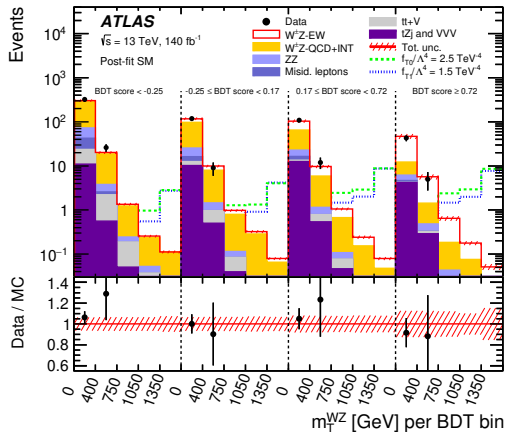
Search for aQGC: Methodology

5 Measurement methodology and results

Anomalous Quartic Gauge Couplings modeled via **EFT dimension-8 operators** drastically enhance the cross-section at high energy scales ($\hat{s}$).

2D Fit Strategy

- **Observables:**
BDT score (EW vs QCD) & m_T^{WZ} .
- **Binning:**
 $4 \text{ BDT} \times 5 m_T^{WZ} \rightarrow \mathbf{1D \text{ histogram (20 bins)}}$.
- **Test Statistic:**
Profiled Likelihood Ratio $q(\vec{c})$ for **95% CL limits** on $f_j^{(8)} / \Lambda^4$.





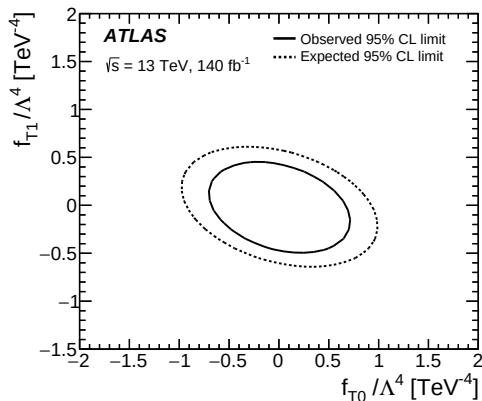
EFT Limits (Without Unitarization)

5 Measurement methodology and results

1D Limits (95% CL)

Coupling	Exp. [TeV^{-4}]	Obs. [TeV^{-4}]
f_{T0}/Λ^4	$[-0.80, 0.80]$	$[-0.57, 0.56]$
f_{T1}/Λ^4	$[-0.52, 0.49]$	$[-0.39, 0.35]$
f_{T2}/Λ^4	$[-1.6, 1.4]$	$[-1.2, 1.0]$
f_{M0}/Λ^4	$[-8.3, 8.3]$	$[-5.8, 5.6]$
f_{M1}/Λ^4	$[-12.3, 12.2]$	$[-8.6, 8.5]$
f_{M7}/Λ^4	$[-16.2, 16.2]$	$[-11.3, 11.3]$
f_{S02}/Λ^4	$[-14.2, 14.2]$	$[-10.4, 10.4]$
f_{S1}/Λ^4	$[-42, 41]$	$[-30, 30]$

2D Contour (f_{T0} vs f_{T1})





Tree-Level Unitarity Violation and Clipping

5 Measurement methodology and results

The EFT is not a complete model. Non-zero dimension-8 operators violate tree-level unitarity at high energies.

The "Clipping" Technique

- EFT contributions are removed above a cut-off scale E_c ($m_{WZ} > E_c$).
- Replaced by SM predictions to restore unitarity.

Unitarized Limits (at crossing bound):

- $f_{T0}/\Lambda^4 \in [-1.5, 1.6] \text{ TeV}^{-4}$
- $f_{T1}/\Lambda^4 \in [-0.7, 0.6] \text{ TeV}^{-4}$

Evolution of 95% CL limits vs cut-off scale $\Rightarrow$

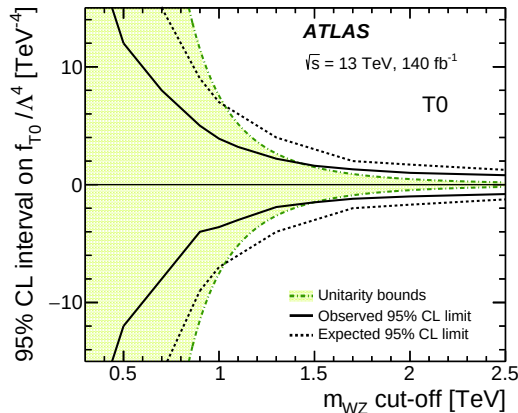




Table of Contents

6 Conclusion

- ▶ Introduction
- ▶ Theoretical Framework
- ▶ Fiducial Phase Space and MC Simulations
- ▶ Event selection & Background estimation
- ▶ Measurement methodology and results
- ▶ Conclusion



Conclusions: VBS as a Precision Probe for the EWK Sector

6 Conclusion

From Discovery to Precision Measurement (2015-2018, 140 fb^{-1})

- **Integrated cross-section:** $\sigma_{WZjj\text{-EW}} = 0.368 \pm 0.037 \text{ (stat)} \pm 0.059 \text{ (syst) fb.}$
- **Differential measurements** (via Bayesian Unfolding) on aQGC-sensitive variables.

Standard Model Verification and Modelling Challenges

- **EW Production:** Excellent Data/Theory agreement in the Signal Region.
- **QCD Background:** Tension of 1.8σ (Data $\sim 30\%$ lower than nominal MadGraph).

Limits on New Physics (EFT) and Unitarity

- **95% CL limits** set on 9 (8) dimension-8 operators without (with) consideration of tree-level unitarity violation.
- These constraints are similar to those obtained by the CMS Collaboration.



Measurements of electroweak $W^{\pm}Z$ boson pair production in association with two jets in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector

Thank you for listening!
Any questions?